Assignment #4 - Facade Pattern

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Topic: Media/File Conversion Facade

1. Introduction

The conversion process has several steps: read the source format, choose a codec, adjust sample data, and combine stereo channels. MediaConversionFacade exposes one convert(fileName, targetFormat) method and runs these steps in order.

The example uses simulated audio samples, as allowed by the assignment. It demonstrates processing without reading or writing real media files.

2. UML Class Diagram

```
Main (Client) ----> MediaConversionFacade
                          |----> MetadataReader.readFormat(...)
                          |----> CodecFactory.createCodec(...)
                          |----> BitrateReader.changeBitrate(...)
                          +----> AudioMixer.mixToMono(...)
```

Each arrow represents a use relationship. The subsystem classes do not refer back to MediaConversionFacade.

3. Clean Code Principles

3.1 Meaningful names

Before: run(a, b) does not explain the operation. After: facade.convert("song.mp3", "wav") names the action and inputs.

3.2 Single Responsibility

Before: one method reads metadata, selects a codec, changes samples, and mixes channels. After: the facade coordinates while each subsystem performs its own task.

3.3 No subsystem details in the client

Before: Main creates and calls four subsystem classes. After: Main calls only MediaConversionFacade.convert(...).

3.4 Small focused classes

MetadataReader reads the format, CodecFactory selects a supported codec, BitrateReader changes sample frames, and AudioMixer combines stereo pairs into mono.

3.5 Clear error handling

Before: an unsupported format could fail later. After: MetadataReader and CodecFactory report invalid input with a clear IllegalArgumentException message, and Main displays it.

4. Conclusion

The facade hides the order of conversion steps and subsystem calls. Advanced clients can still use the public subsystem classes directly. The demo changes simulated samples and prints the result; it does not create an encoded media file.